

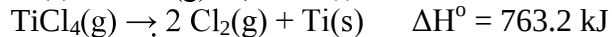
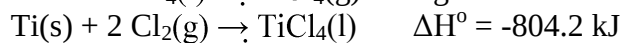
Chapter 5
Thermochemistry

-
1. The specific heat of iron is 0.451 J/g K. What is the molar specific heat of iron?
A. 0.451 J/mol K B. 26.0 J/mol K C. 25.2 J/mol K D. 55.85 J/mol K
-
2. When heat is transferred to the system, the process is said to be ----- and the sign of **q** is -----.
A. exothermic, positive. B. endothermic, positive
C. exothermic, negative D. endothermic, negative
-
3. What is ΔE for a system which has the following two steps?
Step 1: The system absorbs 60 J of heat while 40 J of work are performed on it.
Step 2: The system releases 30 J of heat while doing 70 J of work.
A. 110 J B. 100 J C. zero D. 90 J
-
4. Which of the following is a state function?
A. enthalpy B. work C. heat D. power
-
5. For a particular process $q = 30 \text{ kJ}$ and $w = -25 \text{ kJ}$. What conclusion may be drawn for the process?
A. $\Delta E = 55 \text{ kJ}$
B. the system does work on the surroundings.
C. the heat transfer from the system to the surrounding.
D. the work energy transfer from the surrounding to the system.
-
6. How many joules of heat are required to heat 1.00 g of lead from 25 °C to the melting point (327 °C) and melt all of it? (The specific heat capacity of lead is 0.159 J/g K and it requires 24.7 J/g to convert lead from the solid to the liquid state.)
A. 2.47 J B. 39.4 J C. 48.0 J D. 72.7 J
-
7. When 221 grams of water at a temperature of 25.0 °C is mixed with an unknown mass of water at a temperature of 57.0 °C the final temperature of the resulting mixture is 33.1 °C. What was the mass of the second sample of water?
A. 29.5 g B. 67.0 g C. 74.9 g D. 131 g
-
8. When 108 grams of water at a temperature of 22.5 °C is mixed with 65.1 grams of water at an unknown temperature, the final temperature of the resulting mixture is 47.9 °C. What was the temperature of the other sample of water?
A. 18.9 °C B. 67.0 °C C. 79.7 °C D. 90.0 °C
-
9. Calculate the amount of heat required to heat 357 grams of ethylene glycol (antifreeze) from 0 °C to 110 °C. The specific heat of ethylene glycol is 2.42 J/g K.
A. 16.2 kJ B. 39.3 kJ C. 71.5 kJ D. 95.0 kJ
-
10. Calculate the specific heat of a certain unknown metal, if it requires 195 joules of heat to raise the temperature of 12.1 grams of this metal by 34.6 °C.
A. 0.451 J/g K B. 0.335 J/g K C. 0.714 J/g K D. 0.816 J/g K
-

-
11. If 0.750 g of MgO is placed in a coffee-cup calorimeter and then 100.0 mL of 1.00 M HCl is added, the temperature of the HCl solution increases from 22.8 °C to 28.7 °C. Based on this information, calculate the enthalpy change for this reaction per mol of MgO. You may assume that the specific heat of the solution is 4.20 J/g K and the density of the HCl solution is 1.00 g/mL.
- A. -2.54 kJ/mol B. -29.8 kJ/mol C. -65.3 kJ/mol D. -134 kJ/mol
-
12. When a piece of unknown metal weighing 24.4 g, and at 119.3 °C, is placed in a calorimeter containing 51.0 g of water at 19.6 °C, the temperature increases to 20.8 °C. If the specific heat of the water is 4.18 J/g K and the specific heat of the calorimeter is 159 J/K, what is the specific heat of the unknown metal?
- A. 0.128 J/g K B. 0.186 J/g K C. 0.723 J/g K D. 1.12 J/g K
-
13. The standard molar enthalpy of combustion for C₃H₈(g) is -2044 kJ.
- $$\text{C}_3\text{H}_8(\text{g}) + 5 \text{O}_2(\text{g}) \rightarrow 3 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{l})$$
- What is the standard enthalpy change for the combustion of 3.000 mols of C₃H₈(g)?
- $$3 \text{C}_3\text{H}_8(\text{g}) + 15 \text{O}_2(\text{g}) \rightarrow 9 \text{CO}_2(\text{g}) + 12 \text{H}_2\text{O}(\text{l})$$
- A. -6132 kJ B. -4088 kJ C. -2044 kJ D. +2044 kJ
-
14. What is the value of the molar enthalpy (or heat) of combustion of C₂H₆(g) if the combustion of 3.01 grams of this gas at constant pressure releases 8.47 kilojoules of heat?
- A. -0.847 kJ/mol B. -8.47 kJ/mol C. -84.7 kJ/mol D. -847 kJ/mol
-
15. The molar enthalpy (or heat) of combustion of C₂H₅OH is -277.7 kJ/mol. How many kJ of heat will be released by the combustion of 9.22 g of C₂H₅OH at constant pressure?
- A. 11.8 kJ B. 27.8 kJ C. 32.9 kJ D. 55.5 kJ
-
16. Calculate the standard molar enthalpy of formation for Na₂O(s), given that the standard enthalpy of formation for Na₂O₂(s) is -505 kJ/mol and the enthalpy change for the following reaction
- $$\text{Na}_2\text{O}(\text{s}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{Na}_2\text{O}_2(\text{s}) \quad \text{is } -89.0 \text{ kJ/mol.}$$
- A. 594 kJ/mol B. 416 kJ/mol C. -416 kJ/mol D. -594 kJ/mol
-
17. Calculate the standard molar enthalpy of combustion for ethene, C₂H₄,
- $$\text{C}_2\text{H}_4(\text{g}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$$
- based on the following standard enthalpies of formation: $\Delta H^\circ_f[\text{C}_2\text{H}_4(\text{g})] = +52.26 \text{ kJ/mol}$; $\Delta H^\circ_f[\text{CO}_2(\text{g})] = -393.5 \text{ kJ/mol}$; and $\Delta H^\circ_f[\text{H}_2\text{O}(\text{g})] = -241.8 \text{ kJ/mol}$.
- A. -204.3 kJ B. -583.8 kJ C. -687.6 kJ D. -1322.9 kJ
-
18. The standard molar enthalpy change is -802.3 kJ for the combustion of CH₄(g).
- $$\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$$
- Calculate the standard molar enthalpy of formation for CH₄(g) based on the following standard enthalpies of formation: $\Delta H^\circ_f[\text{CO}_2(\text{g})] = -393.5 \text{ kJ/mol}$ and $\Delta H^\circ_f[\text{H}_2\text{O}(\text{g})] = -241.8 \text{ kJ/mol}$.
- A. -1679 kJ/mol B. -125.4 kJ/mol C. -74.8 kJ/mol D. +892.4 kJ/mol
-

19. Calculate the enthalpy of vaporization for TiCl_4

$\text{TiCl}_4(\text{l}) \rightarrow \text{TiCl}_4(\text{g})$ given the following enthalpies of reaction:



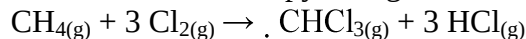
A. -80.4 kJ

B. +41.0 kJ

C. +80.4 kJ

D. +127.3 kJ

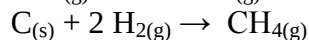
20. Calculate the standard enthalpy change of reaction for the process



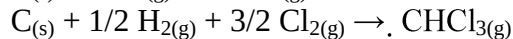
using the following reactions:



$$\Delta H^\circ = +92.3 \text{ kJ}$$



$$\Delta H^\circ = -74.8 \text{ kJ}$$



$$\Delta H^\circ = -103.1 \text{ kJ}$$

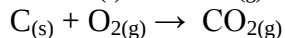
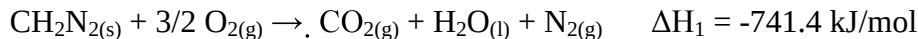
A. -120.6 kJ

B. -212.9 kJ

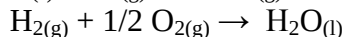
C. -305.2 kJ

D. -454.8 kJ

21. Calculate the standard enthalpy of formation for CH_2N_2 , given the following standard enthalpies of reaction:



$$\Delta H_2 = -393.5$$



$$\Delta H_3 = -285.8$$

A. +62.1 kJ/mol

B. -633 kJ/mol

C. -849 kJ/mol

D. -1420 kJ/mol

ANSWER KEY

Q.	A.	Q.	A.	Q.	A.	Q.	A.	Q.	A.
1	C	6	D	11	D	16	C	21	A
2	B	7	C	12	B	17	D		
3	C	8	D	13	A	18	C		
4	A	9	D	14	C	19	B		
5	B	10	A	15	D	20	C		
